

Kubernetes & AKS Interview Handbook: 120+ Real Interview Questions & Production Scenarios (Beginner → Advanced)

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If you have any questions, get stuck on a concept, or need help with any topic, feel free to DM me on Instagram. I'll do my best to help.

Before You Start

This handbook is not meant to help you **memorize** Kubernetes. It's designed to help you **understand** Kubernetes. Every concept in Kubernetes is connected.

When you understand **Pods**, concepts like **ReplicaSets**, **Deployments**, **Services**, **Ingress**, **HPA**, **StatefulSets**, and **Scheduling** start making sense naturally.

Don't study each topic in isolation. Connect the dots. Think about how Kubernetes works behind the scenes. Once you understand the flow, interviews become conversations instead of question-and-answer sessions.

Section 1 – Kubernetes Fundamentals (Most Asked in Interviews)

1. What is Kubernetes?
2. Explain Kubernetes Architecture.
3. Explain Azure Kubernetes Service (AKS).
4. Explain AKS Architecture.
5. Explain the Kubernetes Platform.
6. What happens internally when you run `kubectl apply -f`?
7. How is a Pod scheduled to a Node?
8. Why do we need Pods when Containers already exist?
9. Difference between Pod, ReplicaSet, and Deployment.
10. What is a Namespace?

Section 2 – Pods & Workloads

11. How is a Pod created?
12. Pod creation journey during HPA.
13. Why is a Pod restarting continuously?
14. OOMKilled in Kubernetes.
15. Existing application works, but after a new deployment Pods fail due to memory. How do you troubleshoot?
16. Application keeps restarting under load. Where will you investigate?
17. Stateless vs Stateful applications.
18. How do you decide whether an application should be Stateful?
19. Can a Stateful application become Stateless?
20. What is a ReplicaSet?
21. What is a Deployment?

22. Why use Deployment instead of directly creating Pods?
23. If a Pod is deleted, will Kubernetes recreate it?
24. Purpose of Job resource.
25. Pod Disruption Budget (PDB).
26. How does PDB work?

Section 3 – Services & Networking

27. Service Types in Kubernetes.
28. ClusterIP.
29. Headless Service.
30. NodePort.
31. LoadBalancer Service.
32. How does NodePort work internally?
33. What is Ingress?
34. Difference between Ingress and LoadBalancer.
35. How Deployment and Service are connected.
36. Service Discovery.
37. Communication within the same Namespace.
38. Communication between different Namespaces.
39. Default communication between Namespaces.
40. Allow only Frontend Namespace to access Backend Namespace.
41. Frontend AKS cannot access Backend AKS – Troubleshooting.
42. Explain end-to-end how external traffic reaches a Pod using Istio in AKS.
43. Istio Gateway vs Istio Ingress Gateway.
44. NGINX Ingress Installation.
45. NGINX Ingress Configuration.

Section 4 – Scheduling

46. How Kubernetes Scheduler works.
47. Deploy application only on selected Nodes.
48. Prevent Pods from scheduling on specific Nodes.
49. Deploy application only on one specific Node.
50. NodeSelector.
51. Node Affinity.
52. Required vs Preferred Node Affinity.
53. Node Affinity YAML.
54. Taints.
55. Tolerations.
56. Taints vs Tolerations.
57. DaemonSet.
58. Static Pods.

Section 5 – Scaling & Resource Management

59. Horizontal Pod Autoscaler (HPA).
60. Primary function of HPA.
61. Configure HPA in Deployment.
62. Configure HPA in AKS.
63. Difference between ReplicaSet and HPA.
64. Can AKS configure HPA automatically?

- 65. Memory utilization keeps increasing even after increasing Node size. How do you solve it?
- 66. Requests vs Limits.
- 67. What happens if Requests are not defined?
- 68. What happens if Limits are not defined?
- 69. ResourceQuota.
- 70. LimitRange.
- 71. Vertical Pod Autoscaler.
- 72. Cluster Autoscaler.

Section 6 – Azure Kubernetes Service (AKS)

- 73. Explain AKS.
- 74. AKS Architecture.
- 75. Node Pool.
- 76. System Node Pool.
- 77. User Node Pool.
- 78. Where do AKS system Pods run?
- 79. Where do application Pods run?
- 80. Azure CNI.
- 81. Azure CNI Overlay.
- 82. Azure CNI vs Kubenet.
- 83. Kubernetes Add-ons.
- 84. kubeconfig file.
- 85. Managed Identity in AKS.
- 86. AKS Networking.

Section 7 – AKS Upgrade & Cluster Maintenance

- 87. Have you upgraded an AKS cluster? <- most asked
- 88. Complete AKS Upgrade Process.
- 89. Control Plane Upgrade.
- 90. Node Pool Upgrade.
- 91. Why upgrade Control Plane first?
- 92. What is kubectl cordon?
- 93. What is kubectl drain?
- 94. What is kubectl uncordon?
- 95. Difference between Cordon, Drain, and Uncordon.
- 96. Rolling Upgrade Process.
- 97. Automatic Upgrade in AKS.
- 98. Production Cluster Upgrade Scenario.

Section 8 – Storage

- 99. Persistent Volume (PV).
- 100. Persistent Volume Claim (PVC).
- 101. StorageClass.
- 102. Dynamic Provisioning.
- 103. ReadWriteOnce vs ReadWriteMany.
- 104. Who creates PVC?
- 105. Persistent vs Ephemeral Storage.
- 106. CSI Driver.
- 107. Database vs Monitoring Storage.

Section 9 – Configuration & Security

- 108. ConfigMap.
- 109. Secret.
- 110. Which component distributes Secrets & ConfigMaps?
- 111. TLS in AKS.
- 112. Certificate Creation Process.
- 113. ServiceAccount.
- 114. RBAC.
- 115. ImagePullSecrets.
- 116. Security Context.
- 117. Pod Security Admission.
- 118. Running Containers as Non-Root.

Section 10 – Multi-Container Pods

- 119. Init Container.
- 120. Sidecar Container.
- 121. Difference between Init Container and Sidecar.
- 122. Waiting for another Service using Init Container.
- 123. Sidecar Dependency.
- 124. CMD-based Dependency.
- 125. InitContainer YAML.
- 126. Sidecar YAML.
- 127. Production Use Cases.

Section 11 – Observability & Troubleshooting

- 128. Application Metrics.
- 129. Infrastructure Metrics.
- 130. Complete Observability Setup.
- 131. Pod Pending.
- 132. ImagePullBackOff.
- 133. ErrImagePull.
- 134. CrashLoopBackOff.
- 135. CreateContainerConfigError.
- 136. CreateContainerError.
- 137. ContainerCreating.
- 138. Evicted Pods.
- 139. OOMKilled.
- 140. High CPU Usage.
- 141. High Memory Usage.
- 142. Pod Logs.
- 143. Fetch Logs from Multiple Pods.
- 144. kubectl describe.
- 145. kubectl get events.
- 146. kubectl top.
- 147. Production Troubleshooting Approach.

Section 12 – Kubernetes Internal Components ()

- 148. kube-apiserver.

- 149. etcd.
- 150. kube-scheduler.
- 151. kube-controller-manager.
- 152. cloud-controller-manager.
- 153. Node Controller.
- 154. Service Controller.
- 155. Route Controller.
- 156. kube-proxy.
- 157. CoreDNS.
- 158. Endpoint & EndpointSlice.
- 159. Labels & Selectors.
- 160. Desired State Reconciliation.
- 161. NetworkPolicy.
- 162. Kubernetes Operators.

Section 13 – Helm

- 163. Helm Architecture.
- 164. Helm Charts.
- 165. values.yaml.
- 166. Override values.yaml.
- 167. Helm Upgrade.
- 168. Helm Rollback.
- 169. Rolling Update Strategy.
- 170. Recreate Strategy.
- 171. maxSurge & maxUnavailable.
- 172. Rollback Deployment.

Section 14 – Most Used kubectl Commands

- 173. kubectl get
- 174. kubectl describe
- 175. kubectl logs
- 176. kubectl exec
- 177. kubectl apply
- 178. kubectl delete
- 179. kubectl replace
- 180. kubectl rollout
- 181. kubectl rollout history
- 182. kubectl rollout undo
- 183. kubectl top
- 184. kubectl cordon
- 185. kubectl drain
- 186. kubectl uncordon
- 187. kubectl get events
- 188. kubectl config